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How drivers' depth perception of environmental features influences traffic speed

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ABSTRACT

Drivers' perception of the traffic environment, particularly their depth perception of surrounding elements, has a significant influence on their driving speed. While street view images (SVIs) capture environmental features from a driver's perspective (such as trees, buildings, and pedestrians), the perceived distance of these elements from the driver (their depth value) plays a critical role in speed regulation that has been understudied. The same features at different distances can create varying levels of psychological pressure on drivers, with closer objects potentially inducing greater caution and lower speeds. To address this research gap, we used a monocular depth estimation model for simulating drivers' perceptions and explored their impact on driving behavior. Specifically, we characterized the traffic conditions across 5458 road segments in Wuhan based on 5 million taxi GPS records. Next, we calculated the depth value from 14,115 panoramic street view images. Using the interpretable machine learning method, we map the spatial distribution of perceived environmental features associated with both low and high driving speeds. Our approach could explain 38.1% of driving speed variations. The results reveal a threshold effect between perceived environmental depth and driving speed, clarifying the non-linear relationship between drivers' subjective perception of spatial distances and their speed choices.

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Driving speed; traffic environments; depth estimate; street view image; explainable machine learning; GeoAI

1. Introduction

As urbanization and population growth continue to accelerate, the challenges faced by urban transport systems become increasingly complex and multifaceted (Kan et al. 2019; Zhang et al. 2023). Among these challenges, traffic congestion stands out as a critical issue that significantly undermines transport efficiency and contributes to a variety of environmental problems, including noise pollution and carbon emissions (Choudhary and Gokhale 2016; Lu et al. 2021; Zhang, Kwan, and Ma 2024). While often attributed to high traffic volume, a significant portion of congestion stems from driver behaviors, such as the reluctance to maintain a consistent and safe driving speed. This hesitation is frequently rooted in the driver's perception and judgment of the surrounding environment. The influence of drivers' perceptions of the traffic environment on driving speed has been widely corroborated in existing research (Casali, Yonca, and Comes 2022). Particularly, misjudging the spatial relationship with surrounding vehicles or objects – a behavior closely linked to depth perception – is associated with drivers slowing unnecessarily. The perceived spatial relationship between drivers and surrounding elements has profound psychological

impacts. For instance, when drivers perceive nearby buildings, pedestrians, or vehicles as being too close, they often report negative psychological experiences, such as stress and anxiety, and are observed to reduce their speed (Ambros et al. 2021; Cai et al. 2022). This phenomenon aligns with theories of environmental psychology, which suggest that a heightened “sense of enclosure” from proximate objects is associated with increased cognitive load and perceived risk, with drivers observed to adopt more cautious behaviors (Akhtar and Moridpour 2021). And this observation highlights a direct link between drivers' depth perception of traffic environments and their driving behaviors, which, in turn, impact the driving speed.

Street view images (SVIs) offer an efficient, wide-reaching and cost-effective method for perceiving urban environments (Biljecki and Ito 2021; Gao et al. 2024), closely aligning with the driver's point of view (Cai et al. 2022). Some researchers have applied the SVIs into simulating drivers' environmental perceptions. However, traditional analyses typically focus on capturing only the two-dimensional characteristics of the visual environment (Bakhtiary et al. 2024; Xu, He, and Jiang 2024). These approaches are limited to extracting semantic segmentation information (e.g.

proportions of trees and buildings) (Qi et al. 2020; Zhang et al. 2021, 2023) or detecting environmental features (e.g. vehicle and pedestrian counts) (Chen et al. 2022; Garrido-Valenzuela et al. 2023).

In reality, drivers' perception encompasses not just what objects are present, but also how far these objects are from them. As shown in Figure 1, even when traffic scenes exhibit similar visual element proportions, the absence of perceptual depth information results in an inaccurate reflection of the driver's subjective experience (Hu et al. 2023). In scenes represented by Figure 1(a,b), elements such as vehicles and trees have similar visual proportions. We cannot determine which scene would result in lower driving speeds merely from the visual proportions or number of vehicles. Yet we cannot adequately estimate which scene would result in lower driving speeds merely from the visual proportions or number of vehicles. When depth information is introduced, in Figure 1(b), the proximity between the driver and other vehicles is noticeably close, which would induce psychological pressure and significantly reduce driving speed.

In summary, it remains unclear how to move beyond simple visual proportion assessment methods to incorporate spatial depth information, explore the visual impact of different elements on drivers, and understand how these effects affect driving speeds. The subjective perceptions of drivers (such as proximity to buildings, pedestrians, and other vehicles (Rahman et al. 2022)) directly affect their psychological comfort and driving behavior, potentially contributing to lower traffic speeds. This study seeks to answer the following research question: How does drivers' perception of the traffic environment, particularly the perceived distances between drivers and different environmental elements (buildings, pedestrians, and vehicles), influence driving speed across various road segments?

The two main contributions of this paper are as follows. (1) We quantified the relationship between drivers' depth perception of traffic environmental

features and driving speed by introducing depth value analysis. This approach goes beyond traditional research, which primarily focuses on two-dimension semantic features. Our method integrated three-dimensional observations, extracted using advanced computer vision techniques, to establish a link between the perceived traffic environment and driving speed (Akhtar and Moridpour 2021; Garrido-Valenzuela et al. 2023). (2) The second contribution is identifying the perceptual comfort zone ranges for different road segments, providing valuable insights for urban design to reduce traffic emissions and congestion. Our research employed an explainable machine learning method to identify the key visual features that locally influence driving speed (Iban 2022; Polson and Sokolov 2017; Liu, Chen, et al., 2023), which allows for the development of tailored solutions for specific road conditions.

The remainder of this paper is structured as follows: Section 2 presents related research work, and Section 3 introduces the methodology, including data collection procedures, depth information extraction from street view images, and the interpretable machine learning approach. Section 4 reports our findings on the relationship between perceived traffic environment features and driving speed. Finally, Section 5 provides conclusions and future directions.

2. Literature review

Driving speed is influenced by multiple factors, such as traffic volume, road design, and public transportation accessibility (Zhang et al. 2025). Assessing the impact of these factors on driving speed is a multifaceted challenge (Rahman et al. 2022; Song et al. 2019). Previous studies have observed that traffic environment features, such as trees or buildings, affect drivers' line of sight, leading to a series of negative psychological experiences such as stress and anxiety (Casali, Yonca, and Comes 2022). Additionally, as the traffic environment becomes more congested, factors such as reduced road width at the physical level or



Figure 1. Similar perceived environments also have different traffic conditions, emphasizing the need for additional information to accurately understand environmental impact on congestion.

increased sense of enclosure (e.g. walls and buildings) at the perceptual level may further affect drivers' driving speeds (Akhtar and Moridpour 2021; Cai et al. 2022).

SVI has proven to be a reliable tool for simulating urban environments and finds widespread application in various domains, including analysis of walkability, access patterns, health sensing, criminal detection, urban poverty, and enabling autonomous driving (Dong et al. 2023; Hankey et al. 2021; Wang et al. 2022; Yuan et al. 2023; Yue, Liu, and Xiao 2023). Taking into account the rich potential of observation from a driver's perspective and large-scale environmental sensing (Hori, Namekawa, and Kanagawa 2020; Zhang, Liu, and Biljecki 2023; Zhang et al. 2018), a considerable number of scholars have applied them to traffic research (Hori, Namekawa, and Kanagawa 2020) (e.g. identification of black spots in traffic [Tanprasert et al. 2020] and traffic safety [Hollander et al. 2021]). They also demonstrated the existence of mutual relationships between drivers' perceived environments and the predictability of driving speed (Jiao and Wang 2023; Tedjopurnomo et al. 2020; Zhang and Raubal 2022), which is the main theoretical foundation of this article.

Quantifying drivers' perception is difficult because human visual perception encompasses not only an understanding of objects (such as shape, color, and size) but also includes multi-dimensional information such as depth and spatial relationships. To simulate the traffic environment, engineers of self-driving vehicles used hardware devices like LiDAR to capture depth information and reconstruct the driving environments (Ma, Xu, and Zhang 2023). For example, impressive studies used depth information to identify trees in the city (Liu, Jiang, et al., 2023) and assess driving safety (Cai et al. 2022). In this research, monocular visual analysis methods were utilized to address this problem.

The application of Explainable Artificial Intelligence (XAI) in the context of traffic has been extensively explored in past research (Gu et al. 2025; Ma et al. 2025). XAI aims to provide transparent and interpretable models and algorithms that can explain how traffic-related decisions are made. Taking advantage of explainable learning (Kim and Kim 2022; Ma et al. 2022), numerous scholars have integrated XAI into transportation analysis (Cui et al. 2025). For instance, Li (2023) employed XAI in conjunction with large-scale travel data to gain insight into the factors that influence the willingness of riders to carpool, providing valuable information on the sustainability and efficiency of urban transport. Further, Liu, Chen, et al. (2023) introduced an interpretable machine-learning framework using panoramic SVI. Through a crowd-sourced scoring approach, they

extracted urban perception measures and employed feature importance and cumulative local effects to enhance interpretability.

In summary, the main goal of this article is to identify which perceived traffic environment features affect driving speed and propose solutions for urban congestion and traffic management (Bao et al. 2022; Chung et al. 2023; Do et al. 2019).

3. Methods and data

This study investigates the relationship between perceived traffic environment and driving speed, explaining how specific environmental features influence driving behavior. We first describe how to extract traffic environment features, especially depth information. Then, we provide the details about the explanation methods and driving speed calculation. This study was carried out on the Wuhan City's third ring road, including 5458 road segments and 14,115 panoramic SVIs (sourced from Baidu Maps, one of the largest online mapping services in China) to simulate drivers' perception of the traffic environment.

3.1. Extraction of drivers' perceived traffic environment from street view images

To simulate drivers' subjective environmental perception based on street view imagery, we employed multiple methods including semantic segmentation and depth estimation. Semantic segmentation is relatively common, aiming to identify different elements within the street scene (such as the visual proportions of trees, vehicles, buildings, etc.). Depth estimation goes further by simulating drivers' spatial distance perception and three-dimensional environmental understanding, which is critical for driving decisions. We introduced the monocular depth scaling (MiDaS) model to estimate the perceived depth distance from SVIs. MiDaS is a real-time monocular depth prediction method designed for RGB images (Ranftl et al. 2020). It used a combination of low-level and high-level features to predict dense depth maps from a single RGB image. This approach enables the model to handle various image content and scale variations effectively.

Monocular depth estimation involves predicting distance information for objects within an image. In contrast to object detection and semantic segmentation, depth estimation provides three-dimensional sensing results that enhance our understanding of the spatial position, shape of objects, and especially the depth distance (Figure 2(c)).

Depth estimation can be conducted using two approaches: one based on depth sensors such as LiDAR or RGB-D cameras, and the other based on monocular models. The latter relies solely on

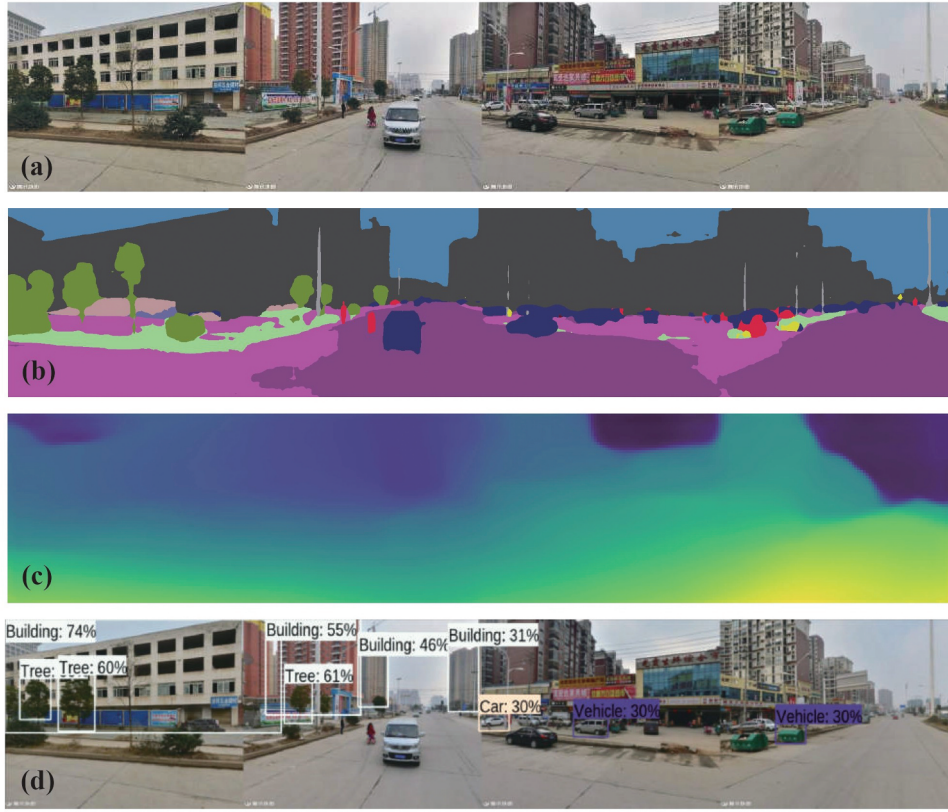


Figure 2. The illustration of simulating drivers' subjective environmental perception based on street view imagery. (a) Original panoramic image. (b) Semantic segmentation results. (c) Depth estimation results. (d) Object detection results.

information from a single image to estimate depth. Monocular depth estimation, which takes a single RGB image as input and generates a depth map, is the most widely applied method (Mertan, Duff, and Unal 2022). The depth map can be further refined by incorporating other information, such as semantic segmentation and geometric constraints. In monocular depth estimation, a per-pixel depth regression task is performed, where a single camera is used to estimate the depth of each pixel in the image. The process is guided by a simple equation for monocular depth estimation:

$$Loss_{MDE} = \frac{1}{N} \sum_{i=1}^N \frac{|z_{gt,i} - z_{pred,i}|}{z_{gt,i}} \quad (1)$$

where $z_{gt,i}$ and $z_{pred,i}$ represent the ground truth and predicted depth values, respectively, at the pixel i , and N is the total number of pixels in the image.

Depth estimation tasks are more challenging compared to image classification tasks. This is because real depth data can come from absolute depth (measured from LiDAR or calibrated stereo cameras) or from disparity maps (from uncalibrated stereo cameras). We do not know the exact depth of each pixel because different cameras have varying focal lengths and imaging qualities. To address these challenges, the MiDaS network proposes a novel combination loss function (compared to Equation (1)) and introduces a depth

scaling layer that learns to predict the scale of the depth map, which is crucial for monocular depth estimation. The loss on dataset I can be calculated using the following equation:

$$\mathcal{L}_I = \frac{1}{N_I} \sum_{n=1}^{N_I} \mathcal{L}_{ssitrim}(\hat{d}^n, (\hat{d}^*)^n) + \alpha \mathcal{L}_{reg}(\hat{d}^n, (\hat{d}^*)^n) \quad (2)$$

where N_I is the size of the training set, and the parameter α is usually set to 0.5 (Ranftl et al. 2020). The loss $\mathcal{L}_{ssitrim}$ represents the scale- and shift-invariant loss, and $\mathcal{L}_{reg}$ is the gradient matching term loss. The losses are evaluated directly on the ground-truth disparity values, while also considering unknown scale and shift. For example, there may be sky regions around tall buildings, and in our common sense, the sky is considered infinitely far away, while the distance to buildings is often within tens of meters. This non-smooth jump can also be handled. This example demonstrates the benefits of using such losses in SVI depth estimation. The predictions can avoid extreme values and non-smoothness. The equation for the scale- and shift-invariant loss $\mathcal{L}_{ssitrim}$ is as follows:

$$\mathcal{L}_{ssitrim}(\hat{d}, \hat{d}^*) = \frac{1}{2M} \sum_{j=1}^{U_m} \rho_{mae}(\hat{d}_j - \hat{d}_j^*) \quad (3)$$

where $|d_j - d_j^*| \leq |d_{(j+1)} - d_{(j+1)}^*|$, and $U_m = 0.8M$ (M represents the total number of pixels in the

image. This parameter helps to mitigate the excessive influence of extreme values on the loss function). $\hat{d}$ and d^* are the aligned predicted and ground truth values. The gradient matching loss L_{reg} (Li and Snively 2018) was calculated as follows:

$$\mathcal{L}_{\text{reg}}(\hat{d}, \hat{d}^*) = \frac{1}{M} \sum_{k=1}^K \sum_{i=1}^M (|\nabla_x R_i^k| + |\nabla_y R_i^k|) \quad (4)$$

where $R_i = d_i - d_i^*$ and R^k represent the differences in the disparity map on the scale k (different depth scales can be used during training). In this study, the monocular depth estimation method was used to calculate the relative distance (i.e. depth) from each pixel in the panorama image to the sampling position of the SVI (the location where the SVI was captured). Subsequently, the average depth for each category was calculated based on the pixel categories obtained from semantic segmentation.

3.2. Localized determination method for the perception comfort zone

Despite the automatic feature learning, extensive data processing capabilities, and robust data fitting offered by deep learning and machine learning, the presence of a large number of parameters often results in a “black box problem” (Barredo-Arrieta, Laña, and Del Ser 2019; Xing and Sieber 2023). Explainable learning presents several advantages in the fields of machine learning, including understanding feature importance, interpreting model predictions, exploring data distribution, and detecting bias and unfairness. The field of transportation, as well as geographic information science, has garnered significant attention in terms of explainable research (Hsu and Li 2023).

Shapley additive explanations (SHAP) is a method commonly used to explain the predictions of machine learning models. Compared to traditional ordinary least squares (OLS) regression, SHAP has several advantages, including the ability to capture non-linear relationships, provide local interpretability, and consider feature interactions (Su et al. 2022). SHAP values can be seen as a measure of feature importance, representing the contribution of a specific feature to the model’s output among all possible feature subsets. Specifically, for a given sample, its SHAP values are a vector where each element corresponds to the SHAP value of each feature in that sample. When explaining the index, various visualization tools can be used, such as SHAP summary plot, SHAP dependence plot, and SHAP waterfall plot (Wei et al. 2023). These tools can help us better understand the model’s prediction results and reveal the contribution of each feature to the predicted outcome, as well as the interactions between features.

The calculation of SHAP values is based on the concept of Shapley values, which quantify the contribution of each feature to the prediction outcome. Given a machine learning model f , with an input vector $x=(x_1, x_2, \dots, x_n)$ and an output scalar $f(x)$. Our objective is to determine the contribution of each feature x_j to the output. SHAP values can be calculated to assess the contribution of each feature. The definition of SHAP values ϕ_i is as follows:

$$\phi_i(v) = \frac{|S|!(|N| - |S| - 1)!}{|N|!} (v(S \cup^i) - v(S)) \quad (5)$$

where x_S represents the sub-vector obtained from x by selecting features from the set S ; N represents the sampling number; $x_S = (x_i)_{(i \in S)}$, $|S|$ denotes the number of elements in the set S ; $f(x_S \cup^j)$ represents the output of the model when the feature x_j is added to the set S , while $f(x_S)$ represents the output of the model with only the features in the set S . By utilizing SHAP values, we can explain the reasons behind machine learning predictions as well as the contribution of each feature to the prediction outcome.

3.3. Estimation the driving speed from GPS trajectory data

It is challenging to accurately measure individuals’ driving speed on a given road segment, as drivers have different driving habits and skill levels. To simplify the problem, we used the mean speed of all vehicles on each road segment to represent the average driving speed. In this study, we collected a dataset of the GPS trajectories and position data of 3802 taxis in 2017 on the third ring road of Wuhan. The dataset has 5,115,512 GPS records. This substantial dataset allowed us to accurately map the average driving speed on 5458 road segments. Figure 3 illustrates how we computed the driving speed using the taxi GPS records. The equation for the calculation process is given below:

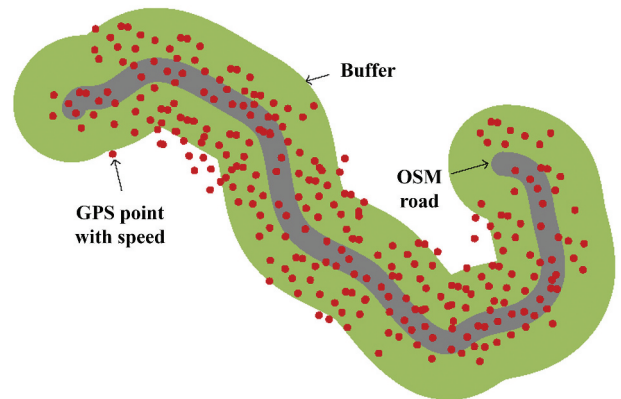


Figure 3. Estimating average driving speed on roads based on taxi GPS data.

$$v_{\text{unit}} = \frac{1}{G^N} \sum_{i=1}^{G^N} v_i \quad (6)$$

where G^N is the number of GPS points within the buffer and v_i represents the individual driving speed at the i th GPS point.

4. Experiment and results

Our analysis consists of two steps, shown in Figure 4. First, we extracted three types of traffic-related data: traffic speed (derived from the GPS records), visual elements of the traffic environment, and depth information (as mentioned in Section 3.1). Second, we comprehensively investigate the relationship between driving speed and perceived traffic environment. This exploration encompassed the application of generalized

least squares (GLS) (Taking into account spatially dependent structures, GLS provides a more efficient parameter estimation than OLS) regression and the Shapley Additive Explanation method (as referenced in Section 3.2). Further, we employed SHAP value maps on a city-wide scale to identify the causes of traffic congestion. As shown in Figure 5(a,b), represent the average driving speed and the daily taxi flow in the study areas, respectively. Figure 5 also provides an overview of six random selected specific research roads, labeled A to F, which will be examined in detail.

4.1. Extraction of drivers' perceived environmental features based on SVIs

SVIs provided a visual representation of the urban environment from a human perspective. As shown in

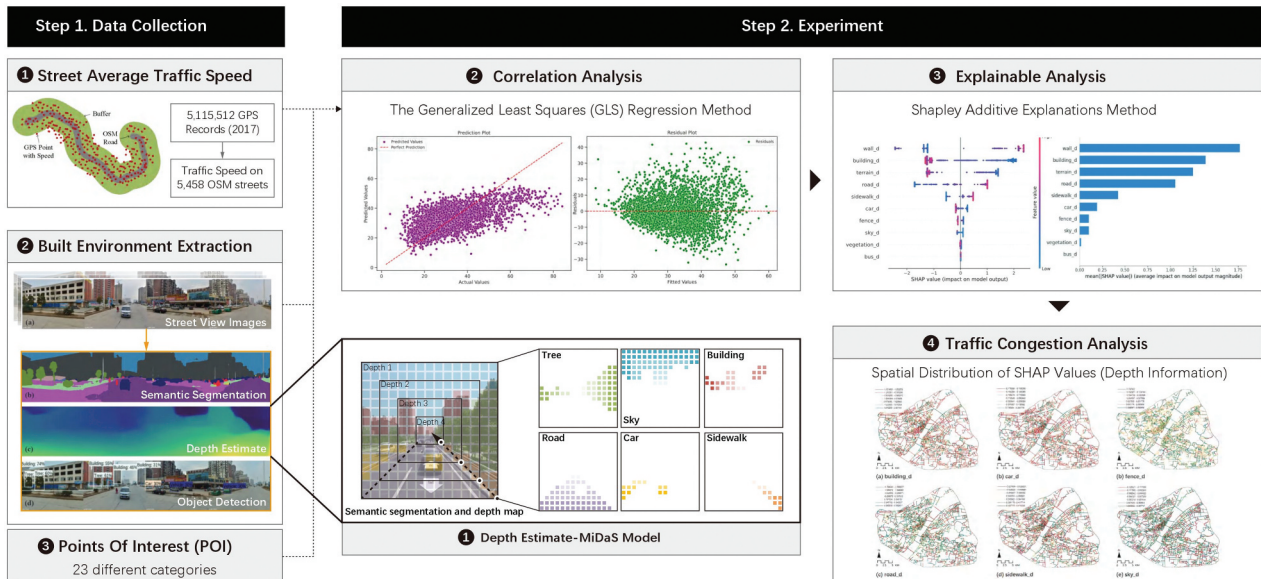


Figure 4. Analytical framework. The research includes two steps. First, we derived three types of traffic-related data: average driving speed for each road segments, visual elements, and depth information extracted from SVIs. Second, we proceeded to the experimental phase, where we comprehensively investigated the relationship between driving speed and perceived environmental features.

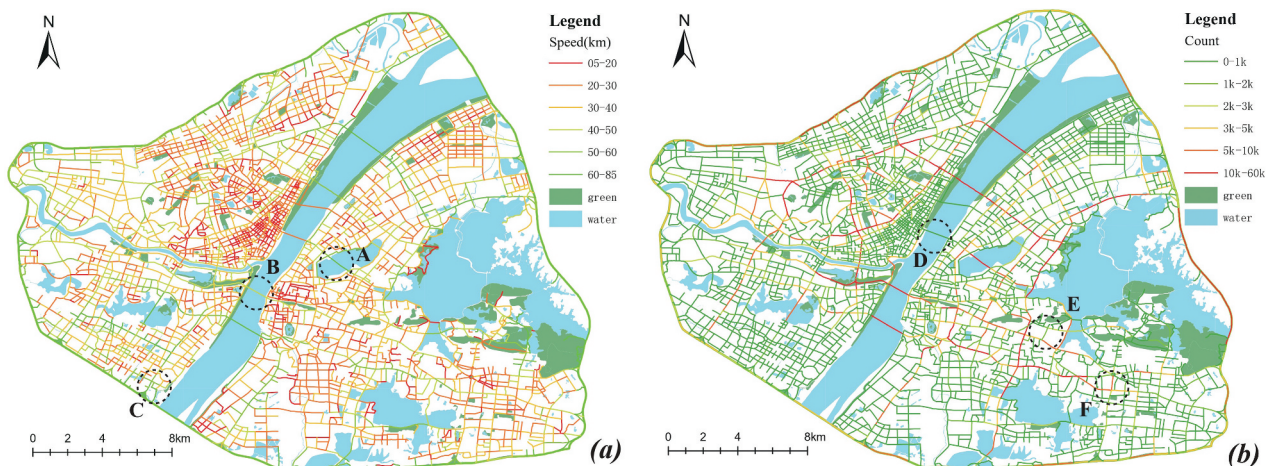


Figure 5. The traffic condition of Wuhan city (July 2017). (a) Average speed and (b) daily taxi flow. (we randomly selected six specific research roads, place a to F, and provide an overview of them, which we will examine in detail).

Figure 2, semantic segmentation was conducted based on the SegNet model, which was pre-trained on the cityscape dataset (including 19 classes) (Badrinarayanan, Kendall, and Cipolla 2017), as an initial step. The feature maps obtained from semantic segmentation were then used to calculate the relative distances between the SVI locations and various elements such as trees, buildings, and cars. Computer vision techniques were used to calculate road features and detect the urban environment. A video demonstration is provided in the Appendix (https://github.com/giserzy/picb/blob/main/depth_view.gif) for a more comprehensive illustration. Table 1 shows the statistical distribution of different proportions of elements in the panoramic images within the study area.

Figure 6 illustrates the process of calculating the relative distances. The semantic segmentation results were used as a mask, and a weighted average of the depth map was computed for each element. It is worth noting that multiple panoramic images may exist for

each OSM road. To simplify the analysis, we took the average attributes of these images to derive the corresponding attribute value for each road.

It is important to acknowledge that the model's predictions are relative values due to uncertainties in scale. For instance, it is not possible to determine the exact distance between drivers and the sky with precision. Additionally, the higher predicted depth values indicate closer proximity to the sampling location, which may initially seem counterintuitive. This is because the model's depth estimation employs an inverse depth representation. For a simple understanding, we can interpret the depth value as a visual pressure value. The average predictions for the depth of each semantic element in the research area are presented in Table 2.

From Table 2, we can observe that the elements closest to SVI locations (with an average relative distance > 20, where a higher value indicates closer proximity and higher visual pressure) are the road surfaces

Table 1. Statistical distribution of different elements of semantic segmentation results in the study streets.

Element	mean	std	min	25%	50%	75%	max
road	0.289	0.067	0.042	0.248	0.297	0.336	0.933
sidewalk	0.023	0.021	0.000	0.008	0.017	0.031	0.185
building	0.198	0.130	0.000	0.102	0.172	0.266	0.771
wall	0.017	0.027	0.000	0.002	0.007	0.020	0.558
fence	0.031	0.031	0.000	0.009	0.023	0.042	0.325
pole	0.002	0.002	0.000	0.001	0.002	0.003	0.029
trafficlight	0.000	0.000	0.000	0.000	0.000	0.000	0.008
trafficsign	0.003	0.005	0.000	0.001	0.002	0.004	0.104
vegetation	0.187	0.135	0.000	0.078	0.163	0.270	0.777
terrain	0.013	0.022	0.000	0.001	0.005	0.016	0.300
sky	0.169	0.106	0.000	0.083	0.158	0.239	0.482
person	0.004	0.006	0.000	0.001	0.002	0.005	0.078
rider	0.001	0.001	0.000	0.000	0.000	0.001	0.012
car	0.038	0.038	0.000	0.009	0.027	0.053	0.471
truck	0.005	0.016	0.000	0.000	0.000	0.002	0.324
bus	0.015	0.036	0.000	0.000	0.003	0.013	0.553
train	0.002	0.009	0.000	0.000	0.000	0.000	0.245
motorcycle	0.003	0.007	0.000	0.000	0.000	0.002	0.125
bicycle	0.001	0.002	0.000	0.000	0.000	0.001	0.039

Note: A higher value indicates a higher proportion (ratio) of the field of view.

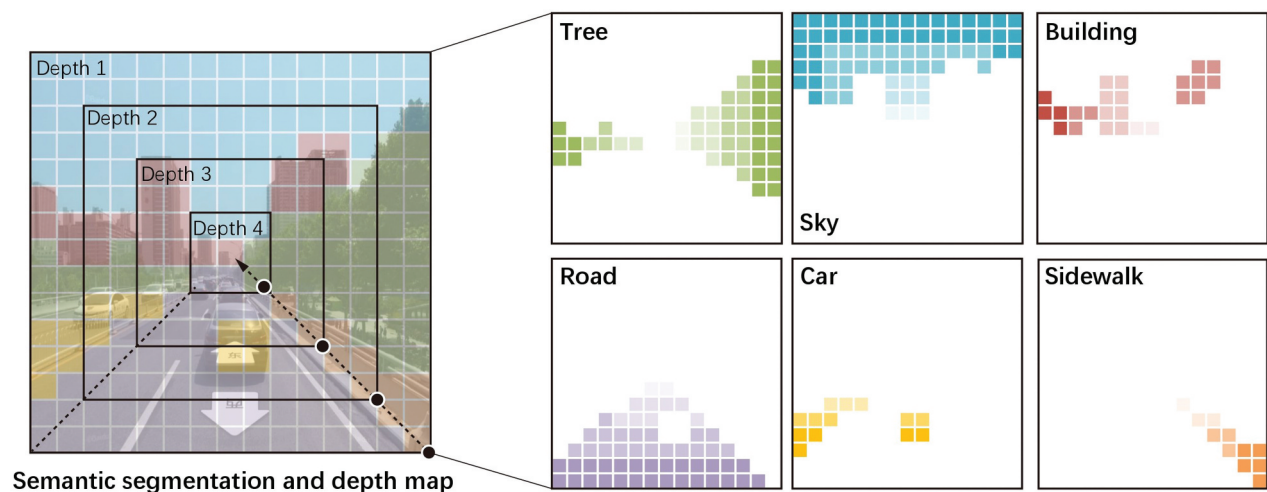


Figure 6. Calculation of relative distances using semantic segmentation and depth map. Each element in the semantic segmentation mask is used to obtain a weighted average depth value. We obtained relative distances, meaning that the further an element is from us, the smaller its depth value. For example, the relative distance of sky elements is 0.

Table 2. Statistical summary of depth measurements for different environmental elements. the table presents minimum, maximum, mean, standard deviation, and quartile values for building depth, vegetation depth, sky depth, road depth, and other physical elements as captured from driver's perspective. These measurements quantify the three-dimensional spatial configuration of street environments, representing the perceived proximity and pressure of different elements to drivers and their potential influence on spatial awareness and driving behavior.

Element	mean	std	min	25%	50%	75%	max
road_d	22.280	3.210	10.116	20.138	22.385	24.483	33.184
sidewalk_d	14.082	3.233	3.538	11.878	13.762	15.985	32.305
building_d	6.274	2.732	0.035	4.477	5.993	7.821	21.072
wall_d	11.817	3.804	1.043	9.339	11.790	13.615	30.919
fence_d	11.089	3.966	0.000	8.352	10.901	13.362	39.071
pole_d	8.732	2.908	0.030	6.848	8.464	10.183	30.208
trafficlight_d	6.745	3.762	-0.474	4.673	6.765	7.265	34.393
trafficsign_d	10.575	4.541	0.000	7.413	9.762	12.850	45.172
vegetation_d	8.239	2.464	0.000	6.703	8.044	9.512	22.695
terrain_d	12.843	3.639	0.646	10.488	12.821	14.653	38.612
sky_d	0.300	0.664	-0.005	0.031	0.077	0.257	13.337
person_d	12.105	4.351	2.533	9.137	11.431	14.429	57.110
rider_d	13.729	5.134	1.395	10.878	13.652	14.805	53.058
car_d	14.993	6.419	0.034	10.666	14.385	18.460	53.504
truck_d	10.663	4.615	0.000	7.794	10.655	11.942	39.129
bus_d	9.886	3.200	0.000	9.063	9.889	9.889	36.065
train_d	11.253	2.650	0.000	11.236	11.236	11.236	39.471
motorcycle_d	16.131	5.396	-0.105	13.025	16.128	17.988	47.777
bicycle_d	15.823	5.787	-0.055	12.442	15.758	18.283	52.559

Note: A higher depth value indicates a closer distance and higher visual pressure, while a lower depth value indicates a lower visual pressure and greater distance to us.

(road). Conversely, the elements that appear farthest are the sky. The depth values for vehicles in the SVI exhibit the highest variance ($\text{std} > 0.6$), indicating a wide range of distances. These observations align with our common understanding and indirectly validate the accuracy of both depth estimation and semantic segmentation.

4.2. Investigating the correlation between perceived environmental features and driving speed

We simulate drivers' perception of the traffic environment based on computer vision techniques, extracting 19 classes of visual features as shown in Section 3.1, including sky, person, rider, car, truck, bus, train, motorcycle, bicycle, road, sidewalk, building, wall, fence, pole, traffic light, traffic sign, vegetation, and terrain. These 19 classes of visual features have been widely applied in areas such as traffic modeling, and are considered to provide a relatively comprehensive perception of the urban traffic&built environment. To explore the impact of perceived environments on driving speed, we first used the GLS regression method to handle the data (encompassing 19 semantic segmentation indicators and corresponding 19 depth feature indicators), which could effectively handle heteroskedasticity and autocorrelation in the data. The GLS model can explain 38.1% of the variance, which means that the perceived environment and traffic speed have considerable associations.

In model fitting, we used robust standard error methods like heteroscedasticity robust (HC3) to deal with heteroscedasticity. As shown in Table 3, the [0.025, 0.975] interval in the regression model

represents the 95% confidence interval for this parameter. In addition, the following legends are used in Table 3 to denote different variables: red dots represent a feature's estimated depth, and yellow dots represent a feature's proportion in the image.

The results indicate that when the distance between the driver and some element (poles, road, and terrain) increases (indicating a wider road, lower depth value, and visual pressure), the psychological pressure on the driver decreases, resulting in faster driving speeds. Contrary to our initial intuition, being closer to walls ($\text{coef} = 0.174$) and vegetation ($\text{coef} = 0.256$) does not impose as much pressure on the driver as we might expect. Regarding walls, considering the prevalence of sound barriers along urban highways, their presence often indicates highway segments, which also do not lead to drivers actively controlling their speed.

Additionally, we generated the prediction plot and residual plot (Figure 7) for the model, which clearly shows a significant association between estimated and actual values of driving speed. However, when the actual driving speed on the road exceeds a certain threshold (approximately 40 km/h), the model's estimated speed does not further increase. It possibly indicates a saturation point beyond which environmental factors have a diminished impact on driving speed.

This section elucidates the complex relationship between drivers' perceived environment features and driving speed, with the results in Figure 7 suggesting that the relationship between these two factors may not be purely linear. In the next section, we will employ machine learning and interpretable methods to model this intricate relationship.

Table 3. GLS regression results showing relationships between perceived environmental features and traffic speed.

R-squared:		0.392	Prob (F-statistic):		0.00		
Model:		GLS	Adj. R-squared:		0.386		
Method:		Least Squares	F-statistic:		62.25		
No. Observations:		3819	Df Residuals:		3782		
Df Model:		36	Covariance Type:		HC3		

Variable	Coef	Std err	t	P> t	[0.025	0.975]	Type
const	−969.7917	119.530	−8.113	0.000	−1204.141	−735.442	●
road_d	−0.2389	0.083	−2.868	0.004	−0.402	−0.076	●
wall_d	0.1842	0.052	3.568	0.000	0.083	0.285	●
pole_d	−0.2577	0.071	−3.608	0.000	−0.398	−0.118	●
traffic_sign_d	0.1140	0.041	2.750	0.006	0.033	0.195	●
vegetation_d	0.2540	0.088	2.878	0.004	0.081	0.427	●
terrain_d	−0.2057	0.049	−4.230	0.000	−0.301	−0.110	●
sky_d	0.7432	0.280	2.656	0.008	0.195	1.292	●
rider_d	0.0837	0.033	2.522	0.012	0.019	0.149	●
car_d	0.1814	0.035	5.180	0.000	0.113	0.250	●
truck_d	−0.0818	0.039	−2.116	0.034	−0.158	−0.006	●
motorcycle_d	0.0930	0.034	2.767	0.006	0.027	0.159	●
bicycle_d	−0.1115	0.030	−3.696	0.000	−0.171	−0.052	●
road	9.9643	1.203	8.281	0.000	7.605	12.323	●
sidewalk	9.4764	1.200	7.899	0.000	7.124	11.829	●
building	10.0467	1.193	8.422	0.000	7.708	12.386	●
wall	10.2915	1.198	8.592	0.000	7.943	12.640	●
fence	10.1659	1.195	8.507	0.000	7.823	12.509	●
pole	12.3382	1.509	8.174	0.000	9.379	15.298	●
traffic sign	8.3254	1.266	6.575	0.000	5.843	10.808	●
vegetation	9.9919	1.194	8.371	0.000	7.652	12.332	●
terrain	10.1871	1.195	8.528	0.000	7.845	12.529	●
sky	10.4726	1.193	8.781	0.000	8.134	12.811	●
person	8.6920	1.305	6.661	0.000	6.134	11.250	●
car	9.5707	1.202	7.960	0.000	7.213	11.928	●
truck	10.1653	1.204	8.445	0.000	7.805	12.525	●
bus	9.9822	1.199	8.327	0.000	7.632	12.333	●
train	10.8471	1.198	9.055	0.000	8.498	13.196	●
motorcycle	9.5455	1.247	7.655	0.000	7.101	11.990	●
bicycle	10.2884	1.506	6.831	0.000	7.335	13.241	●
Catering Services	−0.0894	0.018	−4.889	0.000	−0.125	−0.054	●
Scenic Spots	−0.9102	0.179	−5.085	0.000	−1.261	−0.559	●
Road Facilities	0.1253	0.059	2.122	0.034	0.010	0.241	●
Lifestyle Services	−0.0649	0.019	−3.345	0.001	−0.103	−0.027	●
Sports and Leisure	−0.1443	0.070	−2.074	0.038	−0.281	−0.008	●
Automobile Sales	−0.4351	0.185	−2.346	0.019	−0.799	−0.071	●
Automobile Services	1.2070	0.487	2.479	0.013	0.252	2.162	●

Omnibus:	326.927	Durbin-Watson:	1.573
Prob(Omnibus):	0.000	Jarque-Bera (JB):	504.469
Skew:	0.654	Prob(JB):	2.86×10^{-110}
Kurtosis:	4.207	Cond. No.	6.03×10^4

4.3. Explainable analysis between perceived environment features and traffic flow

In this section, we integrate both machine learning models (we selected XGBoost here) and SHAP values to assess the relationship between drivers' perceived environment and driving speed (Ke, Zhao, and Zhao 2025). This method has been shown to effectively assess the spatial contributions of different traffic

environment features. Importantly, when compared to traditional spatial regression methods, it excels in handling non-linear and complex relationships (Li 2022).

To simplify the calculation and avoid random errors, we filtered out deep features with an average pixel proportion higher than 1% based on the urban elements shown in Table 1. The SHAP summary graph is shown on the left of Figure 8, while the right one represents the

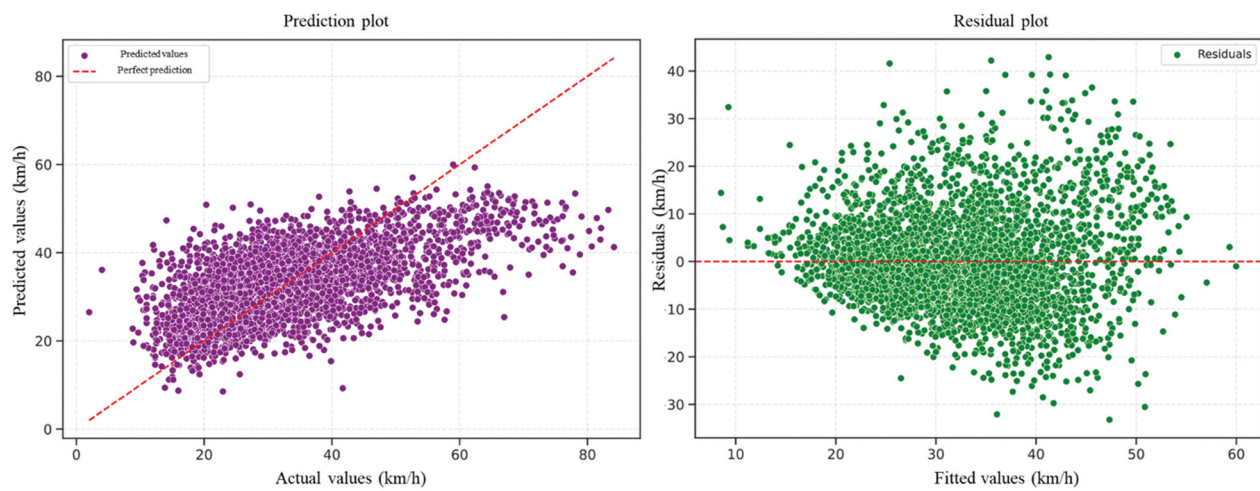


Figure 7. Model performance evaluation for GLS regression of driving speed. Left panel: scatter plot comparing observed driving speeds against model-predicted speeds across all street segments on Wuhan's third ring road, with the diagonal line representing perfect prediction. Right panel: residual plot showing prediction errors against predicted values, with reference line at zero.

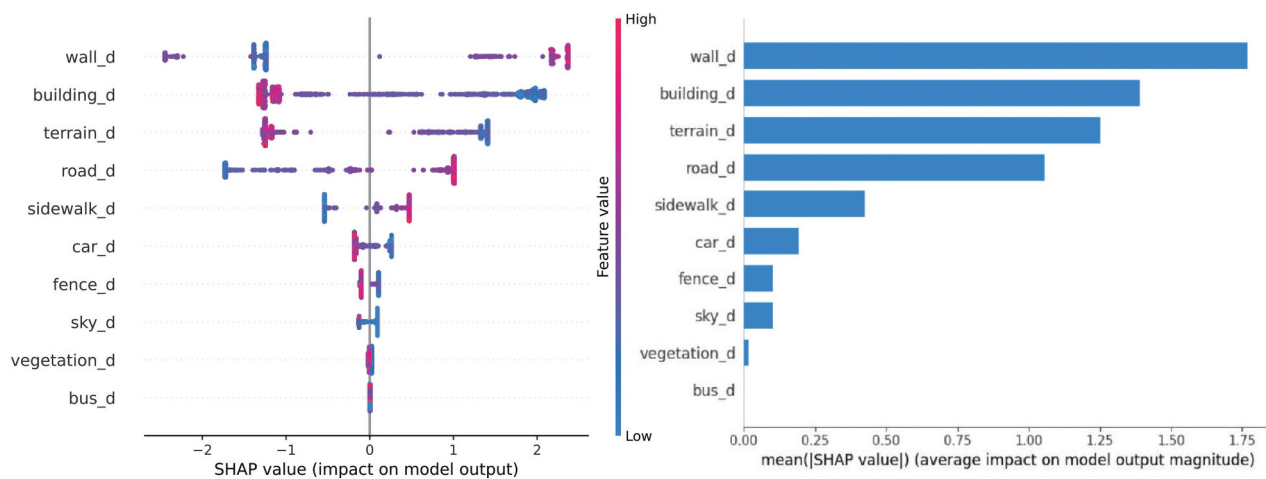


Figure 8. The overall contribution of each perceived environment feature to the model's predicted driving speed (shap summary plot). For the left figure, the x-axis represents the range of shap values for each feature, the y-axis represents the feature names, and each point represents the average shap value of that feature across all samples. This plot helps identify the features with larger or smaller contributions to the model's output and allows for comparison of the relative importance of different features.

feature importance. The y-axis represents the feature ranking based on the sum of SHAP values for all samples, while the x-axis represents the SHAP values, indicating the distribution of the impact of each feature on the model output. On the left panel, each point corresponds to a road segment, and the samples are stacked vertically. The color of each point represents the feature value, with red indicating high values and blue indicating low values. Taking the second row as an example, it suggests that a high value of the feature “building_d” (red color, indicating a closer distance and higher visual pressure) has a negative impact on driving speed. This implies that driving close to a building may affect the driver's behavior and then result in a speed decreasing.

In addition, to gain a deeper understanding of the complex relationship between drivers' perceived traffic environments and driving speed, we used the results on the importance of features (Figure 8). We selected the

six most significant traffic environment depth features that influence the speed: walls, buildings, roads, terrain, sidewalks, and cars, and generated SHAP dependence plots for these elements (see Figure 9). We observed that among these features, their impact on driving speed exhibits sudden shifts from positive to negative (or from negative to positive) as depth changes of entities. It appears that when certain distance thresholds are exceeded (Threshold Effects), their perceptual impact on drivers suddenly changes or intensifies. From Figure 9, it can be seen that maintaining a certain distance from buildings (value < 5), terrain (value < 12), and cars (value < 14) is necessary to create a relatively comfortable perceptual comfort zone. In contrast, for elements such as wall (value > 12), road (value > 21), and sidewalk (value > 14), higher visual pressure (higher depth value and closer distance) produces a positive effect on driving speed. The common

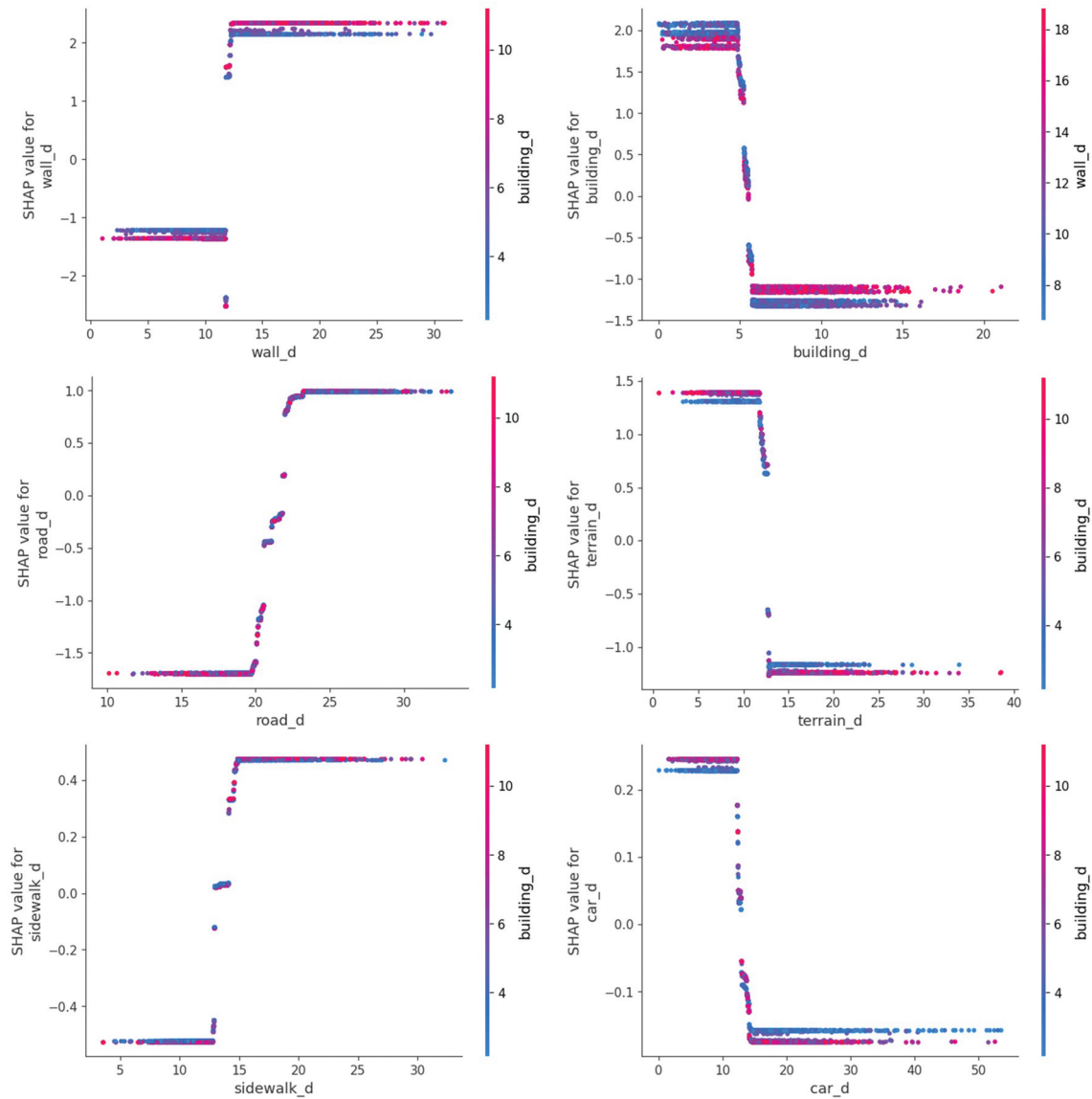


Figure 9. Shap dependence plots showing the threshold effect between perceived traffic environment features and driving speed. The figure illustrates the contribution of each perceived environment feature to the estimation results and the interactions between features in the XGBoost model.

presence of sound barriers on urban highways raised the importance of the wall element. Overall, in road design, we need to consider not only the proportion of different elements in urban areas but also the distance to these different elements, which can create a more comfortable driving environment.

4.4. Traffic congestion spatial analysis with shap value

To further investigate the spatial patterns of traffic speed (i.e. traffic congestion), we selected six typical locations, as shown in Figure 5, and utilized the SHAP waterfall plots (Figure 10) to comprehend the contribution of each environmental feature toward driving speed, providing interpretations at the road segment level. The x -axis represents the cumulative contribution of the SHAP value for each traffic environment feature

to the driving speed. The y -axis represents the name of each feature, and the length of each bar represents the degree of contribution. In Figure 10, $E[f(x)]$ represents the baseline value of the feature, which is the model's prediction of the target variable when no features are taken into account. On the other hand, $f(x)$ represents the probability that the model estimates the proper driving speed for a given road segment. A positive SHAP value indicates that the corresponding feature value increases the estimated speed value, while a negative SHAP value indicates the opposite.

For example, in the case of road A (Located in Figure 10), the building_d (value = 0.875) has the most substantial positive effect on the predicted driving speed. As shown in Table 2, the mean value of building_d is 6.274. Considering that the depth value is relative to road A, the distance from the building is greater than the average distance (lower visual pressure). This deviation

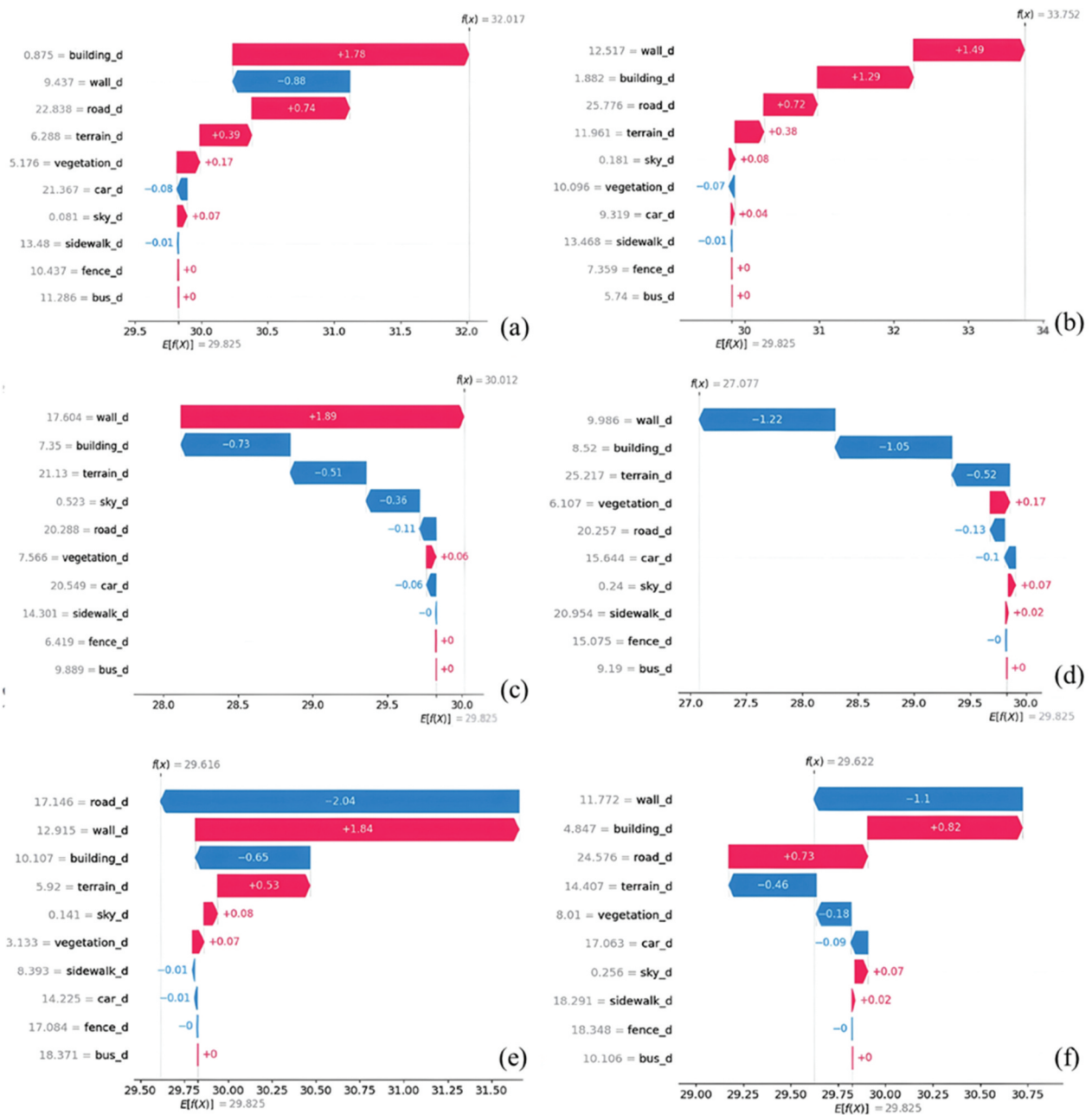


Figure 10. Shap waterfall plots demonstrating the feature-by-feature breakdown of driving speed predictions for six representative road segments (A-F) along Wuhan's third ring road. The X-axis indicated the importance of different features that impacted traffic speed on each road, while the Y-axis represented the contribution degree of each feature. Subplots (a)-(f) represent data from road segments A-F respectively.

from the average has a positive effect on the predicted driving speed. In contrast, in the case of road B, the building_d (value = 8.52, which is higher than the mean value) has a negative effect on the predicted driving speed. The phenomenon is easily explained, as shown in Figure 9, the threshold value for building_d is 5, and its impact on speed is non-linear.

This observed threshold effect represents a critical psychological transition point where buildings begin to dominate a driver's peripheral vision, creating a "wall effect" that heightens the sense of spatial confinement

and triggers more cautious driving behaviors with reduced speeds, altering both depth perception and speed judgment.

In addition to facilitating the identification of the contribution of traffic environment features, we visualized the spatial distribution of SHAP values for all perceived environment features using a red-green bivariate scheme (Figure 11). It reflects both the direction and magnitude of these features' impact on predicted traffic speed. In Figure 11, green lines indicate that the environmental element makes a positive

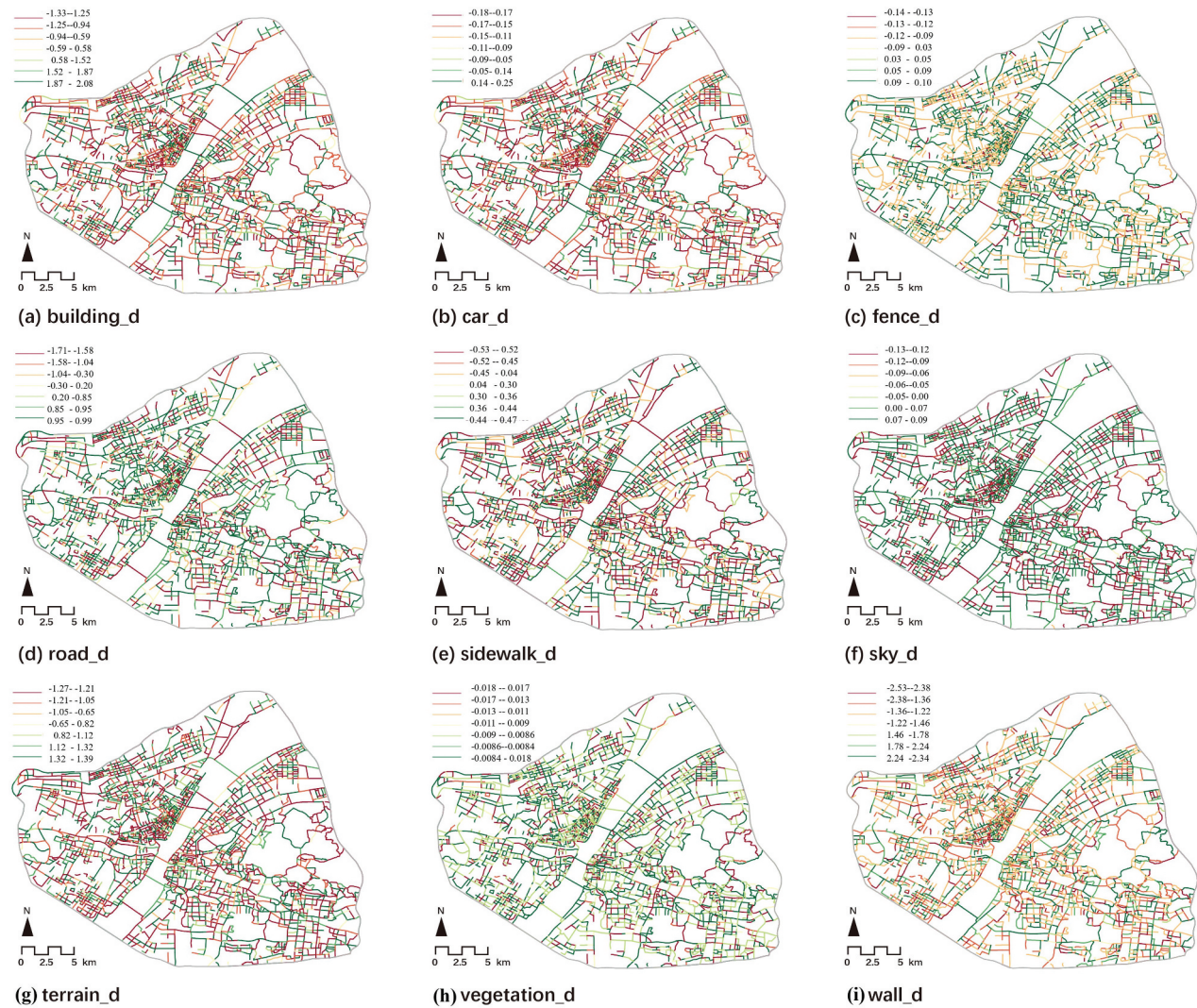


Figure 11. Shap value spatial distribution for depth features' contribution to driving speed. Negative contributions are represented by red, while positive contributions are represented by green. The intensity of the color indicates the magnitude of the contribution. This plot aggregates the waterfall results across all road samples to visualize the spatial heterogeneity.

contribution to the predicted driving speed on that specific road segment. Conversely, red lines indicate that the element has a negative contribution to the predicted speed on that segment. For example, the map for building depth (building_d) shows many green lines in certain areas, indicating that in these specific areas, the building depth feature positively influences the predicted driving speed. Additionally, the distribution map for sky depth (sky_d) shows a large concentration of red lines in certain areas, indicating that in these areas, the sky depth feature negatively impacts predicted driving speed.

Overall, the impact of these depth features on driving speed is generally consistent with previous experiments in Section 4.2. However, there are slight differences on a larger spatial scale. For example, as seen in Figure 11(a), on most roads, the greater the distance between a building to the driver, the higher the predicted traffic speed. But there are a few roads where the effect is the opposite. The reasons for this have been systematically discussed in the threshold effect in the

previous section, suggesting that the influence of environmental features on driving speed is often non-linear and may vary depending on the specific road segment.

5. Discussion and conclusions

This study employed XAI and SVI to examine the influence of drivers' perceived traffic environment on driving speed. By focusing on the depth perception of visual features such as walls, buildings, roads, terrain, sidewalks, and cars, we uncover intricate relationships between these traffic environment features and driving speed. Our approach integrates visual and depth perception analyses to quantify how perceived distances and traffic environment features affect people's driving behaviors.

One significant finding is the identification of a threshold effect, which occurs when the perceived distance of certain traffic environment features surpasses specific thresholds, altering their impact on driving speed. These threshold effects highlight a complex, non-linear relationship between drivers' perceptions

and traffic conditions. Additionally, we generated spatial effect maps to visualize and analyze the spatial patterns of these threshold effects, offering a clearer understanding of how different elements in the drivers' visual field contribute to variations in traffic speed.

This research takes a significant step forward in understanding the role of drivers' perceived traffic environment in driving speed and congestion. We observed that drivers' perceived environmental features explain approximately 38.1% of the variance in road traffic speed. The discovery of specific distance thresholds and their nonlinear effects on driving speed underscores the complex interplay between drivers' perceived environments and actual traffic conditions. For example, it could enable precision interventions. Selective building setback adjustments in congested segments can reduce psychological pressure without comprehensive infrastructure reconstruction. Strategic tree placement can exploit the vegetation proximity threshold by positioning mature trees at distances that provide psychological comfort while maintaining sight lines. The wall proximity threshold suggests that temporary acoustic barriers can be strategically deployed during construction periods to prevent congestion, while portable planters positioned at optimal distances can achieve immediate traffic calming effects. In addition, our road surface depth insights enable precision pavement interventions where high-contrast lane markings can psychologically expand perceived road width in constrained segments. There exists a "perceptual comfort zone" for drivers, within which they feel most comfortable and maintain optimal driving speeds, leading to more efficient and less congested traffic systems.

However, the study faces limitations, including the use of relative depth values, which may not precisely reflect the actual distances affecting drivers' perceptions. Additionally, the findings are based on the specific case of Wuhan's third ring road, and their generalizability requires further examination. Urban morphology, for instance, varies significantly between cities; the "canyon effect" created by dense, tall buildings in cities like Hong Kong or New York might induce a stronger sense of enclosure and thus have a more pronounced impact on speed compared to the environment in Wuhan. Furthermore, regional driving cultures and norms could influence how drivers perceive and react to their surroundings. Therefore, future research should validate these findings across more diverse urban forms and cultural contexts to establish a more universally applicable theoretical framework.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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